

Dharshan Sundarraj

+91 70109 49388 | dharshan.sundarraj@gmail.com | Cuddalore, Tamil Nadu

<https://www.linkedin.com/in/dharshansundarraj/>

<https://github.com/DharshanSundarraj>

<https://dharshansundarraj.github.io/Portfolio/>

Professional Summary

Information Technology graduate and Full-Stack Java Developer with demonstrated experience building core system offline-resilient web applications, software development-ready ACID-compliant REST APIs, and full-stack IoT systems. Proficient in Spring Boot, MySQL, and JavaScript; comfortable across the complete development lifecycle from database schema design to responsive UI delivery. Seeking an entry-level software engineering role where strong Java foundations, real-world project delivery, and Agile collaboration skills can contribute to scalable backend systems driven by passion and clear responsibility for quality.

Technical Skills

Languages	Core Java, JavaScript (ES6+), HTML5, CSS3, SQL.
Backend & APIs	Spring Boot, REST APIs, Java Database Connectivity (JDBC), MySQL, Maven
Databases	MySQL, relational schema design, query optimisation, ACID compliance
Frontend	Responsive UI, PWA (LocalStorage, Service Workers, Web APIs), DOM manipulation
Tools & Technologies	Git/GitHub, VS Code, Figma, Arduino IDE, Agile/Scrum methodology
Core Competencies	Data Structures & Algorithms (DSA), OOP, Unit Testing (JUnit), Software Design Patterns

Technical Projects

TransitPay Ecosystem: Offline-Resilient Transit Ledger | Lead Developer

- **Tech Stack:** Java, Spring Boot, MySQL, Vanilla JavaScript, HTML/CSS, PWA (LocalStorage, Web APIs), html5-qrcode.
- **Offline-First Architecture:** Engineered a Progressive Web App using navigator.onLine and LocalStorage APIs to queue 500+ fare transactions locally during network outages, achieving seamless zero-loss sync upon reconnection.
- **ACID-Compliant Backend:** Architected a Spring Boot REST API with strict @Transactional boundaries and structured exception handling, ensuring exact fare deductions against an immutable MySQL ledger across concurrent sessions.
- **QR Scan Integration:** Integrated html5-qrcode library to enable contactless fare logging in under 2 seconds per scan, reducing manual entry errors to zero.

IoT Smart Authentication System | Lead Full-Stack Developer (Final Year Academic Project)

- **Tech Stack:** Java, MySQL, HTML/CSS/JavaScript, Arduino IDE, ESP32-CAM.
- **Full-Stack Architecture:** Led the complete software development lifecycle for a 4-person team, engineering the end-to-end data pipeline from ESP32-CAM facial recognition capture through a Java server to a live administrative dashboard.
- **Server & Database:** Designed a normalised MySQL schema and built a centralised Java server to securely process, store, and retrieve facial recognition events with sub-1.2-second door-trigger latency.
- **Hardware-Software Integration:** Programmed the ESP32-CAM microcontroller via Arduino IDE to trigger a solenoid lock exclusively upon successful server-side validation, eliminating false-access events.

Digital Ecosystem Redesign (TNSTC & Paper Glass)

- **Tech Stack:** Figma, UI/UX Design, Component Architecture.
- Redesigned the mobile application interface for the Tamil Nadu State Transport Corporation, optimizing user navigation and accessibility.
- Developed "Paper Glass," a comprehensive, responsive UI component library utilizing advanced auto-layout and prototyping logic for future frontend handoffs.

Work Experience

Frontend Development Intern Codveda Technologies

(Virtual) | Dec 2025 – Jan 2026

- Collaborated in virtual environments to translate design mockups into functional, responsive web components for a core system user experience.
- Owned responsibility for implementing UI components with clean, maintainable code and participated in structured code reviews and remote agile workflows.

IoT intern Power Integrated

(Virtual) | May 2025 – Jun 2025

- Developed embedded C programming skills for microcontrollers to interact with temperature, humidity, and light sensors.
- Designed IoT systems for real-time data collection and processing, gaining practical insight into system connectivity.

Education

B.E. in Information Technology - Annamalai University, Cuddalore, Tamil Nadu

OGPA: 7.24 / 10

Relevant coursework: Data Structures, DBMS, Computer Networks, Software Engineering

Certifications

- **UI/UX Wireframing and Prototyping with Figma** – SkillUp / Coursera (Mar 2026)
- **Oracle Java Foundations** – Oracle / Coursera (Mar 2026)
- **Foundations of User Experience (UX) Design** – Google / Coursera (Dec 2025)
- **Introduction to Industry 4.0 and Industrial IoT** – NPTEL (Oct 2025)
- **Privacy and Security in Online Social Media** – NPTEL (Oct 2025)